**KPI Definition Sheet**

1. Introduction

This document defines the Key Performance Indicators (KPIs) used in the Healthcare Operations & Financial Intelligence Dashboard.

The KPIs are designed to measure hospital encounter activity, patient utilisation, encounter duration, procedure utilisation, healthcare costs and payer coverage.

2. Encounter and Patient KPIs

2.1 Total Encounters

**Definition**: Measures the total number of unique healthcare encounters recorded in the hospital database.

Calculation: Counts the distinct encounter IDs in the encounters table.

DAX Measure:
Total Encounters =
DISTINCTCOUNT(encounters[Id])

Business purpose: Helps management understand the overall volume of healthcare services delivered.

2.2 Unique Patients

Definition: Measures the number of distinct patients associated with recorded encounters.

Calculation:Counts distinct patient IDs in the encounters table.

DAX Measure:
Unique Patients =
DISTINCTCOUNT(encounters[PATIENT]).

2.3 Total Patients

Definition:Measures the total number of unique patient records available in the patients table.

Calculation: Counts distinct patient IDs.

DAX Measure:
Total Patients =
DISTINCTCOUNT(patients[Id])

3. Healthcare Cost KPIs

3.1 Total Claim Costs

Definition: Measures the sum of all recorded encounter claim costs.

Calculation: Adds the values in the TOTAL_CLAIM_COST column.

DAX Measure:
Total Claim Costs =
SUM(encounters[TOTAL_CLAIM_COST])

3.2 Average Claim Cost

Definition: Measures the average claim cost per encounter.

Calculation:Divides the total recorded claim costs by the number of encounters with valid claim-cost values.

DAX Measure:
Average Claim Cost =
AVERAGE(encounters[TOTAL_CLAIM_COST])

3.3 Total Base Encounter Costs

Definition: Measures the total recorded base encounter costs.

Calculation:Adds all values in the BASE_ENCOUNTER_COST column.

DAX Measure:
Total Base Encounter Costs =
SUM(encounters[BASE_ENCOUNTER_COST])

4. Encounter Duration KPIs

4.1 Average Encounter Duration

Definition:Measures the average duration of encounters with valid start and stop timestamps.

Calculation: Calculates the elapsed time between START and STOP for valid encounters, then calculates the average in hours.

4.4 Encounters Over 24 Hours

Definition:Measures the number of valid encounters lasting more than 24 hours.

4.5 Encounters Over 24 Hours Percentage

Definition: Measures the proportion of all encounters that lasted more than 24 hours.

Calculation: Divides encounters lasting more than 24 hours by total encounters.

5. Payer Coverage KPIs

5.1 Total Payer Coverage

Definition: Measures the total recorded amount of payer coverage across all encounters.

Calculation:Adds all values in the PAYER_COVERAGE column.

5.2 Average Payer Coverage

Definition: Measures the average payer coverage amount per encounter with a recorded value.

5.3 Zero Coverage Encounters

Definition: Measures the number of encounters where payer coverage is recorded as zero.

5.4 Zero Coverage Percentage

Definition: Measures the percentage of all encounters with zero recorded payer coverage.

DAX Measure:

```
Zero Coverage Percentage =
DIVIDE(
    [Zero Coverage Encounters],
    [Total Encounters],
    0
)
```

5.5 Payer Coverage to Claim Cost Percentage

Definition:Measures total recorded payer coverage as a proportion of total recorded claim costs.

## 6. Procedure Utilisation KPIs

### 6.1 Total Procedures

Definition: Measures the total number of procedure records in the procedures table.

### 6.2 Unique Procedure Types

Definition: Measures the number of distinct procedure codes recorded in the dataset.

### 6.3 Average Procedure Base Cost

Definition: Measures the average recorded base cost of procedures.

DAX Measure:
Average Procedure Base Cost =
AVERAGE(procedures[BASE_COST])

### 6.4 Total Procedure Base Costs

Definition: Measures the sum of all recorded procedure base costs.

Business purpose: Provides an overview of the total recorded base costs associated with procedures.

### 6.5 Procedure Frequency

Definition: Measures how frequently each procedure appears in the dataset.

Calculation: Counts procedure records grouped by procedure description or procedure code.